Car Price Analysis Report

1. Project Overview

This project focuses on analyzing car pricing trends using a dataset of 2500 cars. The goal is to understand pricing distribution across brands, models, and other attributes. The project uses Python for data cleaning, SQL for analysis, and Power BI for visualization to derive meaningful business insights.

2. Dataset Summary

The dataset contains 2500 records with features such as brand, model, price, year, fuel type, transmission, and condition. The dataset is well structured and evenly distributed across brands.

3. Exploratory Data Analysis using Python

Data preprocessing was performed using Pandas. Missing values were handled, duplicate records were removed, and new features such as price categories were created. The cleaned dataset was then exported to SQL for further analysis.

4. Data Analysis using SQL

1. Total Cars & Average Price

This query calculates the total number of cars and the average price, giving an overview of the dataset scale and pricing level.

SELECT count(car_id), round(avg(price),2) FROM car_details;

Result Grid | Filter Rows:

total_no_of_cars	avg_price
2500	52638.02

Insight: The dataset contains 2500 cars with an average price around 52K, indicating mid-range market positioning.

2. Cars per Brand

This query identifies how many cars are available for each brand, helping understand dataset distribution.

SELECT brand, count(car_id) FROM car_details GROUP BY brand;

Insight: Brands are evenly distributed, ensuring unbiased analysis.

Result Grid | Filter Rows:

brand	total_cars
Tesla	348
BMW	358
Audi	368
Ford	347
Honda	352

3. Average Price by Brand

This query compares average pricing across brands to identify premium and budget segments.

SELECT brand, round(avg(price),2) FROM car_details GROUP BY brand;

Result Grid | Filter Rows:

brand	avg_price
Tesla	53475.55
BMW	54157.11
Audi	51953.42
Ford	51593.25
Honda	52050.28
Mercedes	53191.09
Toyota	52078.73

Insight: BMW and Tesla appear as premium brands, while Ford is relatively affordable.

4. Top 5 Most Expensive Cars

This query identifies the highest priced cars in the dataset.

SELECT brand, model, price FROM car_details ORDER BY price DESC LIMIT 5;

Result Grid | Filter Rows:

brand	model	price
Audi	A3	99982.59
BMW	3 Series	99968.62
Tesla	Model S	99905.9
Tesla	Model 3	99794.46
Mercedes	E-Class	99754.42

Insight: Most expensive cars are close to 100K, showing clustering in premium segment.

5. Year-wise Price Trend

This query analyzes price trends over years to identify patterns.

SELECT year, round(avg(price),2) FROM car_details GROUP BY year;

Result Grid | Filter Rows

year	avg_price
2000	53094.16
2001	48944.06
2002	57197.48
2003	52260.19
2004	51517.98
2005	57534.32
2006	53437.6
2007	56464.2
2008	53104.79
2009	52932.74
2010	53650.24
2011	51414.87
2012	57092.14
2013	52495.03
2014	54202.58
2015	51516.46
2016	51977.46

Insight: Prices fluctuate but remain stable overall.

6. Top 3 Cars per Brand.

```
with carss as (select
        brand,
    model,
    price,
        row_number() over(partition by brand order by price desc ) as rnnk
    from car_details
)
select brand,model,price
from carss
where rnnk <=3
order by  brand , rnnk ;
```

Result Grid | Filter Rows:

brand	model	price
Audi	A3	99982.59
Audi	A3	99541.32
Audi	Q5	98434.45
BMW	3 Series	99968.62
BMW	3 Series	99371.68
BMW	5 Series	99073.74
Ford	Focus	99605.33
Ford	Fiesta	99153.11
Ford	Explorer	98716.27
Honda	CR-V	99578.74
Honda	Civic	99500.97
Honda	Accord	99212.85
Merc...	E-Class	99754.42
Merc...	E-Class	99159.41
Merc...	C-Class	99072.6
Tesla	Model S	99905.9
Tesla	Model 3	99794.46
Tesla	Model S	99496.42
Toyota	Camry	99400.47
Toyota	RAV4	98645.75
Toyota	RAV4	98496.67

Insight: Each brand offers multiple high-value models.

18. Cars Above Brand Average

This query counts cars priced above their brand average.

```
WITH avg_table AS (
    SELECT brand, AVG(price) AS avg_price
    FROM car_details
    GROUP BY brand
)
SELECT
    c.brand,
    COUNT(CASE WHEN c.price > a.avg_price THEN 1 END) AS above_avg_count
FROM car_details c
JOIN avg_table a
ON c.brand = a.brand
```

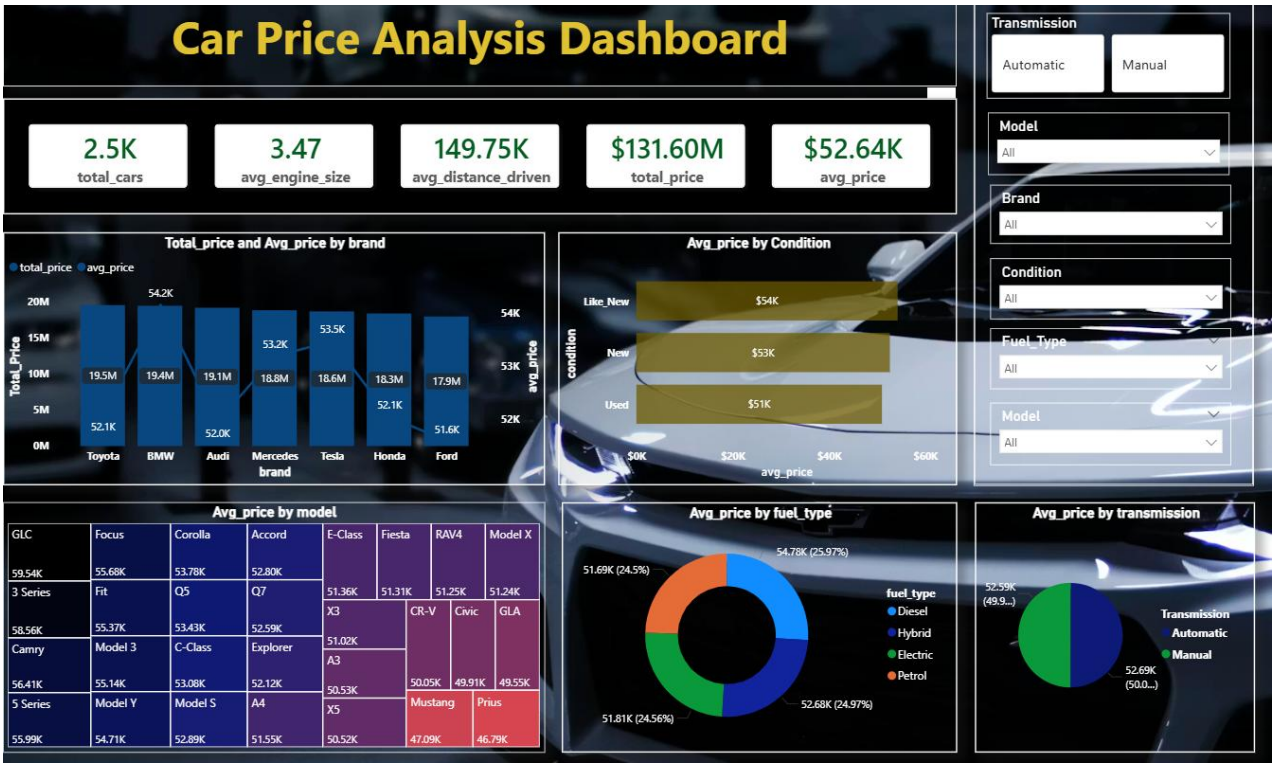
GROUP BY c.brand;

brand	above_avg_count
Tesla	176
BMW	200
Audi	178
Ford	172
Honda	179
Mercedes	177
Toyota	187

Insight: Some brands have more premium offerings than others.

5. Dashboard in Power BI

An interactive dashboard was created to visualize insights such as average price by brand, fuel type, condition, and transmission. It provides a comprehensive view of pricing trends and helps in decision making.



6. Business Recommendations

• Focus marketing on premium brands like BMW and Tesla. • Optimize pricing for mid-range cars. • Analyze high-value models for better positioning. • Use insights for targeted customer segmentation.